

Plant transport: follow the water and sugar

Teacher guide and worked answers

Explain plant transport and use potometer evidence to infer how air movement affects water uptake.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C37

Describe plant transport & transpiration.

Directly develops C37 and Pearson 6.7-6.10,6.12-6.13. Core routes explain transport and simple rate calculations; optional volume calculation extends mathematical demand without assuming Higher-tier entry.

Prior learning

Photosynthesis makes glucose using light energy.

Water can move across partially permeable membranes by osmosis.

Rate is change divided by time.

Success evidence

Distinguish xylem water/mineral transport from phloem sucrose transport.

Explain evaporation and diffusion of water vapour at a leaf.

Calculate a potometer rate and explain why uptake is a proxy for transpiration.

Equipment

Printed shared evidence and one selected response route; pen, ruler and calculator.

Projector or offline HTML/PowerPoint. Optional word processor or spreadsheet.

Optional prepared demonstration: school-approved potometer with leafy shoot, stable support, timer, ruler, thermometer and cool fan. Complete paper route requires none of these.

Preparation

Print the shared evidence page, one chosen response route and exit page. Routes are alternatives, not a workload ladder.

Read the teacher answers before teaching. All numerical sample data in this lesson are constructed for learning.

If demonstrating, teacher/technician assembles and leak-tests the apparatus; cuts and inserts the shoot under water to avoid air entering the xylem. Pupils do not force stems into glass tubing.

Compare the same shoot and capillary over equal intervals; allow adjustment after changing conditions. Monitor temperature/light and keep them comparable. Repeat conditions and record actual results separately.

Practical care

Default task is paper/data analysis. Optional potometer must be prepared and checked by teacher/technician.

Adults handle sharp cutting and fragile glass. Secure the apparatus, keep water away from electrical equipment and use a stable cool fan.

No tasting plant material; wash hands after handling. Use a plant approved for school handling and avoid known irritants.

Ready to teach alternative

Use the supplied diagram, repeated movement table and source cards. Draw arrows and calculate the same rates without assembling a potometer.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	How does water reach a leaf?
2	Retrieval	2-5	What moves and why?
3	I do	5-9	I follow one material at a time
4	I do	9-12	The leaf links water loss to transport
5	I do	12-15	I name the measured quantity
6	We do	15-18	Air-movement investigation
7	We do	18-20	Trace the explanation backwards
8	We do	20-22	Repair the instrument claim
9	Hinge	22-25	Choose the accurate transport statement
10	You do	25-26	Prepare the transport report
11	You do	26-36	Make the explanation traceable
12	Review	36-38	Audience: follow my arrows
13	Review	38-39	A careful transport explanation
14	Exit	39-40	Name the path and the proxy
15	Exit	Within stage	Choose the next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening How does water reach a leaf?

Invite prediction; no need to know transpiration term yet. Show the planned distinction between material transport and a measurement of uptake.

2 Retrieval What moves and why?

R1 glucose. R2 osmosis. R3 change/amount divided by time. Explain glucose can be converted to sucrose for transport; root uptake is not sugar absorption from soil.

3 I do I follow one material at a time

Model blue path from roots to leaf. Root hairs give large area; water can enter by osmosis, while mineral ions may require active transport. Then model leaf-to-bud and leaf-to-root sucrose paths. Xylem conducting cells are hollow/dead with lignified walls; phloem includes living cells and energy-dependent loading/transport processes.

4 I do The leaf links water loss to transport

Think aloud: "First liquid becomes vapour: evaporation. Then vapour spreads out through the pore: diffusion. I should not call these the same process." Wind can remove humid air near the leaf, increasing the vapour gradient if other conditions are comparable.

5 I do I name the measured quantity

Trace the bubble towards shoot. "This is movement along a capillary, not water vapour counted at a leaf. Most uptake may replace loss, but some water is used/retained; leaks matter." Equal tubing permits comparative movement rates; volume requires cross-sectional area.

6 We do Air-movement investigation

Still mean 12 mm; moving mean 30 mm. Rates 2.4 and 6 mm/min. Constructed matched shoot/light/temperature/capillary data; effects are conditional, not a universal fixed wind multiplier.

7 We do Trace the explanation backwards

Moving air removes humid air near stomata, increasing diffusion gradient; more water evaporates/is lost; uptake through xylem tends to replace it. No claim fan pushes water through leaf. Accept accurate arrow chain.

8 We do Repair the instrument claim

"Water-uptake proxy" or "bubble movement indicating uptake." Uptake may differ from loss; volume not directly known from distance alone. Ask what extra quantity converts distance to volume: cross-sectional area.

Reveal: Measure bubble movement per time. Infer water uptake; treat transpiration as a proxy relationship.

9 Hinge Choose the accurate transport statement

B correct. A swaps tissue/material; C swaps measurement/inference; D misses source/sink.

A. Xylem carries sucrose made in the roots. - Xylem carries water/mineral ions; phloem transports sucrose between sources and sinks.

B. Water evaporates inside a leaf and vapour diffuses through stomata. - Yes: state change first, then diffusion through the pore.

C. A potometer directly measures all water vapour leaving a leaf. - A bubble potometer measures uptake; it does not collect all vapour loss.

D. Phloem transport is always downward. - A sink may be above or below a source; different tubes can transport in different directions.

10 You do Prepare the transport report

Shared evidence supplied. Let pupils point or narrate arrows rather than require lengthy prose.

11 You do Make the explanation traceable

Supported S1-S4; standard M1-M4; stretch T1-T4. Check that moving air explanation includes vapour gradient, not fan pushing liquid. If numbers confuse, model rate from a 10 mm/5 min example.

12 Review Audience: follow my arrows

Quiet partner/TA review. Pupil adds tissue label or changes “exact transpiration” to an accurate uptake label.

13 Review A careful transport explanation

Model still 2.4 versus moving 6 mm/min in the supplied setup; stronger air movement is consistent with faster uptake, not a direct measurement of every evaporated molecule.

14 Exit Name the path and the proxy

E1 phloem. E2=4 mm/min. E3 some water used/retained and apparatus limitations. Collect one precise response.

15 Exit Choose the next useful step

Shared exit minute. Revisit a source leaf with sinks above/below if downward-only misconception persists.

Answers and assessment

Retrieval

R1 glucose, which can be converted to sucrose for transport. R2 osmosis. R3 change/amount divided by time.

W1-W3

W1 means 12 mm and 30 mm; rates $12/5=2.4$ mm/min and $30/5=6$ mm/min. W2 moving air can remove humid air, increasing the vapour gradient and loss; xylem uptake tends to replace that loss. W3 bubble movement indicates uptake, not exact evaporation; distance alone also lacks volume unless area is known.

Hinge

A false: xylem/materials reversed. B correct: evaporation then diffusion. C false: uptake measurement is a transpiration proxy. D false: phloem transports source-to-sink, which can be up or down in different tubes.

S1 / M1

Xylem: water/mineral ions, hollow dead conducting cells with lignified walls, mainly root-to-leaf stream. Phloem: living cells, dissolved sucrose from sources to sinks; different tubes can serve sinks above/below. Energy is involved in phloem transport processes.

S2 / M2

Water enters root cells by osmosis and moves through root tissues into xylem. Water is drawn along xylem towards leaves; it evaporates from moist internal surfaces and water vapour diffuses through stomata. Root hairs increase uptake area; mineral-ion uptake may require active transport.

S3 / M3

Still mean $(12+11+13)/3=12$ mm; moving mean $(30+28+32)/3=30$ mm. Divide each by 5 minutes: 2.4 and 6 mm/min. Difference 3.6 mm/min; moving value 2.5 times still in this constructed setup.

S4

The bubble tracks water entering the shoot. Some water is used in processes or retained, and water stores can change; leaks affect readings. Thus uptake and loss are related but not exactly identical.

M4

Moving air can remove humid air near stomata and increase the diffusion gradient, promoting water loss and replacement uptake. Same shoot/leaf area, capillary, light, temperature and time; allow adjustment, repeat, and check seal/leaks. Any two justified controls/checks earn the method element.

T1

Still: $0.5 \times 12 = 6 \text{ mm}^3$ over 5 minutes; $6/5 = 1.2 \text{ mm}^3/\text{min}$. Moving: $0.5 \times 30 = 15 \text{ mm}^3$; $15/5 = 3 \text{ mm}^3/\text{min}$. Units must distinguish volume from distance.

T2

The same distance represents different volumes when cross-sectional areas differ. Compare volume per time or use the same capillary; do not treat mm/min as universally comparable uptake.

T3

No. Strong water stress can close stomata, reducing the diffusion pathway and water loss. Environmental effects depend on plant response; the original moderate-condition trend is not unlimited.

T4

Apparatus: inspect/verify an airtight seal or test for leaks with an appropriate prepared setup. Biology: repeat with matched shoots/leaf areas or repeat condition order while controlling temperature/light. Accept an explicit check that targets an identified source of uncertainty.

Exit and assessment

E1 phloem. E2 $20/5 = 4 \text{ mm}^3/\text{min}$. E3 water used/retained or stores changing; a relevant apparatus limit also shows caution. L1 accepts a specific improvement or justified confirmation. Secure core includes correct tissue/material, evaporation/diffusion distinction and proxy-aware rate interpretation.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 10, We do 7, hinge 3, independent 11, review 3, exit 1. Zero-minute slides share their stage time.

Choose one response route. Accept accurate speech, pointing plus explanation, annotations or writing as evidence.

A quiet review still has an audience: teacher or TA reads the pupil's decision, asks one evidence question, and the pupil revises or confirms it.

The phloem diagram shows different transport paths; it does not claim simultaneous opposite bulk flow in one sieve tube. Keep source and sink tied to organ roles, which can change over a plant's life.

Interactive We do predicts the effect of moderate moving air in the exact constructed table, tests the calculated rates, then checks the uptake/transpiration distinction. Paper W1-W3 uses the same values. Animated bubble motion is controlled and represents a measurement model, not a real plant simulation.

Access and responsive teaching

Supported

Same core scientific goal, with word bank, short prompts, read-aloud or pointing/dictation. Do not assign a route from diagnosis.

Standard

Use the supplied evidence to explain and justify an independent scientific decision.

Stretch

Optional challenge after secure core: evaluate a limitation and transfer the science. Higher-tier-only material, where present, is labelled explicitly.

Regulation

Preview the sequence. Offer solo or partner work, a pass and quiet response. Allow a brief reset and re-entry at the next numbered task; no compulsory disclosure, performance or speed competition.

Misconceptions to check

Xylem carries plant food from the roots.

Xylem carries water and mineral ions; photosynthesis makes organic material and phloem moves sucrose.

Check: Ask which material came from photosynthesis.

Phloem only flows down.

Sucrose moves from a source to a sink; different tubes can carry it in different directions.

Check: Point to a growing bud above a source leaf and roots below it.

Liquid water simply escapes as liquid through stomata.

Water evaporates from internal cell surfaces; water vapour diffuses through stomata.

Check: Ask pupils to name the state change and then the movement process.

Potometer uptake equals exact transpiration.

Much absorbed water replaces transpiration loss, but some is used or retained; leaks and storage changes also affect readings.

Check: Ask what quantity the bubble movement measures directly.

Word	Meaning
xylem	Transport tissue with hollow, lignified dead conducting cells carrying water and mineral ions.
phloem	Living transport tissue that moves dissolved sucrose between sources and sinks.
transpiration	Loss of water vapour from aerial plant parts, mainly through leaf stomata.
translocation	Transport of dissolved organic substances such as sucrose in phloem.
stoma	A pore in the leaf surface, controlled by guard cells.
potometer	Apparatus that measures water uptake by a shoot; uptake can indicate transpiration.

Scientific references

[Pearson Edexcel GCSE Biology 1B10 specification](#)

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 8 September 2026.

[SAPS: Investigating transpiration with a potometer](#)

Checked water-uptake measurement and the distinction from mass-loss methods. All movement values in this lesson are constructed. Checked 8 September 2026.

OpenStax Biology 2e: Transport of water and solutes in plants

Checked xylem/phloem, source-to-sink transport, stomata and water movement. Checked 8 September 2026.